Chapter 16: Waves

Continuous Mass with Continuous Motion

Background

Continuum mechanics encompasses systems that can not be described as point masses or as rigid bodies such as the systems treated in earlier chapters of Taylor. When the mass of a system is distributed continuously and the movement of it's relative coordinates is also continuous, we can use waves to describe our system. Continuous waves appear in many vastly different physical systems. Waves can describe the motion of fluids like water and air, seismic shaking that travels through the earth's crust, or the motion of a taught surface perturbed from rest. In the next section we will consider the latter case as we examine a Gaussian perturbation of a taught 2D surface with boundary conditions. First, however, we will need to build up a basic understanding of wave mechanics.

For simplicity, we will derive the one-dimensional wave equation using the same process as Taylor and then generalize the result to two dimensions for our problem. To arrive at the 1D wave equation, we will consider a linear continuous distribution of mass (i.e. a string) that is displaced from equilibrium. In this case, all of the motion of the string is in the y-direction. We should stop here and make some simplifying approximations. Since we want to work with a constant value for the string's tension T, we should assume that the displacement is very small so that the strings length does not change much. Our system can also be broken up into very small line segments connected by point masses. This is approximation is visualized in Fig. 1.

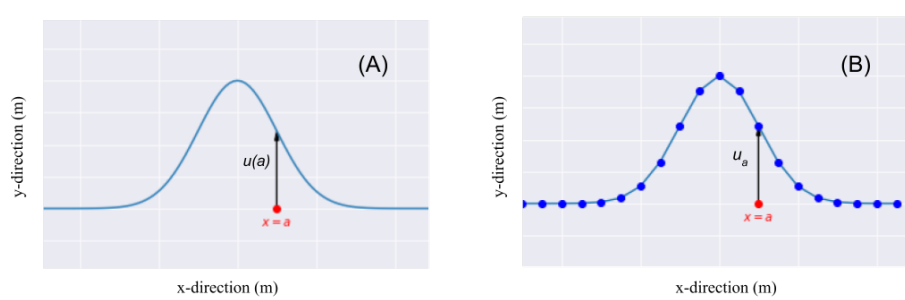

Figure 1: Plot (A) shows the perturbed string as a continuous distrubution of mass. Plot (B) shows the approximation of the string as a collection of line segments, connected by point masses. In both cases, the displacements from equilibrium $u(a)$ and u_a is shown at the location $x = a$.

For our string system, we want to write an equation of motion that describes the disturbance of each line segment from equilibrium at time t. For a true con-

tinuous distribution, this is be given by a function $u(x,t)$. We will approximate u as the vertical displacement of a small line segment that stretches from x to $x+dx$. We will first derive the force F_{net} from tension on one of these line segments. Then Newton's second law allows us to equate F_{net} to the mass of each line segment μdx times its acceleration $\frac{\partial^2 u}{\partial t^2}$. F_{net} is simply the sum of the tension forces from each adjacent line segments. Assume the angles between a line segment and its two adjacent neighbors are ϕ and $\phi + d\phi$ respectively. Then trigonometric relations tell us that the x and y components of the tension force on the line segment with tension T are given by,

$$F_{net,x} = -Tcos(\phi) + Tcos(\phi + d\phi), \text{ and} \tag{1}$$

$$F_{net,y} = -Tsin(\phi) + Tsin(\phi + d\phi) = Tcos(\phi)d\phi. \tag{2}$$

As expected for purely vertical motion, we see that the right hand side of equation (1) disappears as angles ϕ and $d\phi$ approach zero. Equation (2), on the other hand, reduces to a phase shifted cosine which approaches 1 as ϕ gets smaller. We can now replace $cos(\phi) \approx 1$ with $\frac{dx}{dx}$ and find that F_{net} reduces to,

$$F_{net,y} = T\frac{\partial \phi}{\partial x}dx, \tag{3}$$

where we have switched to the partial derivative notation since ϕ also depends on time. We can finally approximate the small angle ϕ as the slope of the line segment to replace ϕ in equation (3) with $\frac{\partial u}{\partial x}$. Now $F_{net,y}$ is just the second partial derivative of u with respect to x times dx. As prefaced, we equate this to $\mu dx \frac{\partial^2 u}{\partial t^2}$ (the mass times acceleration of the segment) leaving us with,

$$\frac{\partial^2 u}{\partial t^2} = \frac{\mu}{T}\frac{\partial^2 u}{\partial x^2}. \tag{4}$$

The constant $\sqrt{\frac{\mu}{T}}$ is the wave speed commonly denoted v in wave mechanics. Now we have derived the one-dimensional wave equation given by equation (4). To describe the motion of a taught *surface*, we will need to use the 2D form of equation (4). The derivation in two dimensions is analogous, and the result is unsurprisingly given by,

$$\frac{\partial^2 u}{\partial t^2} = \frac{\mu}{T}\Big(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\Big). \tag{5}$$

Before moving on to our problem solution, we leave the reader with one final note on solutions to the wave equations and their relation to normal modes. Despite new math results, the chain of point masses connected by string segments is physically reminiscent of a coupled oscillator. Just as the coupled oscillator has a set of normal modes and normal frequencies that describe periodic motion, so should the wave equation. We will find that for continuous mass distributions with infinitely many degrees of freedom, the set of normal modes is infinite. We can (and shall soon) use this infinite set of normal modes to construct particular solutions to the wave equation. The normal modes and frequencies of a continuous system are standing waves defined by the region's boundaries.

Problem Solution

Now that we understand where the wave equation comes from, we can find a particular solutions from a set of boundary conditions and an initial disturbance from equilibrium. We will consider a taut surface with dimensions that measure 3m by 3m. The surface is under tension such that it has a wave speed of 0.5 m/s, and the edges of the membrane are fixed to have a wave amplitude of 0. The entire membrane starts at rest and with a Gaussian perturbation in the surface, given by,

$$u(x, y, 0) = 5e^{-4(x-1)^2-4(y-1)^2}. \tag{6}$$

By solving the 2D wave equation, we should be able to describe the height of the surface at any location (x, y) and any time t.

Part A: Guess at a Solutiom

Recall that the 2D wave equation is given by,

$$\frac{\partial^2 u}{\partial t^2} = v^2\Big(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\Big), \tag{7}$$

where v is the wave speed and u is the displacement of the height from equilibrium. We are going to solve (7) by separation of variable, so we should guess at a solution that is a product of independent contributions from each variable. A solution of this form can be written as,

$$u(x, y, t) = X(x)Y(y)T(t). \tag{8}$$

The first step is to plug our guess solution (8) into the 2D wave equation (7). Doing so yields the following,

$$X(x)Y(y)\frac{\partial^2}{\partial t^2}T(t) = v^2T(t)\Big(Y(y)\frac{\partial^2}{\partial x^2}X(x) + X(x)\frac{\partial^2}{\partial y^2}Y(y)\Big), \tag{9}$$

where we have factored out independent terms from each partial derivative. Now we can divide the entirety of equation (9) by $X(x)Y(y)T(t)$ to isolate the time dependent terms on the left. This leaves us with

$$\frac{1}{T(t)}\frac{\partial^2}{\partial t^2}T(t) = v^2\Big(\frac{1}{X(x)}\frac{\partial^2}{\partial x^2}X(x) + \frac{1}{Y(y)}\frac{\partial^2}{\partial y^2}Y(y)\Big). \tag{10}$$

Part B: Writing the Most General Solution

Next, we can make some conclusions about the terms on each side of equation (10). We realize that since the right hand side is independent of t and the left hand side independent of x and y, this means that each side must be equal to a constant. Any variance of the left hand side with t would break the equality in (10), since the right hand side *cannot* change with t.

Since we know each side of equation (10) is a constant, we can write the following two equations

$$\frac{1}{T(t)}\frac{\partial^2}{\partial t^2}T(t) = const, \text{ and} \tag{11}$$

$$v^2\Big(\frac{1}{X(x)}\frac{\partial^2}{\partial x^2}X(x) + \frac{1}{Y(y)}\frac{\partial^2}{\partial y^2}Y(y)\Big) = const. \tag{12}$$

Moving the x dependent part in (12) to the right side, we can make the same argument about each side's respective independence from the other to conclude that these terms must also be constant. Naming our three constant terms a, b, and c, we can now write the following three (single variable) differential equations:

$$\frac{1}{T(t)}\frac{\partial^2}{\partial t^2}T(t) = a, \tag{13}$$

$$\frac{1}{X(x)}\frac{\partial^2}{\partial x^2}X(x) = \frac{b}{v^2}, \tag{14}$$

$$\frac{1}{Y(y)}\frac{\partial^2}{\partial y^2}Y(y) = \frac{c}{v^2}. \tag{15}$$

Remembering that the terms on the left hand side come from equation (10), we can easily deduce that $a = b + c$. Now the differential equations given in equations $(13) - (15)$ are familiar, so we can immediately write down their most general solutions as follows,

$$T(t) = A_1 cos\Big((b+c)t\Big) + A_2 sin\Big((b+c)t\Big), \tag{16}$$

$$X(x) = B_1 cos\Big(\frac{b}{v^2}x\Big) + B_2 sin\Big(\frac{b}{v^2}x\Big), \tag{17}$$

$$Y(y) = C_1 cos\Big(\frac{c}{v^2}y\Big) + C_2 sin\Big(\frac{c}{v^2}y\Big), \tag{18}$$

Part C: Boundary Conditions for x and y

With the most general solution in hand, it is time to start finding some constant from the boundary conditions. We will right down our four spatial boundary conditions now, given by (i)−(iv).

(i) $u = 0$ at $x = 0$	(ii) $u = 0$ at $x = 3$
(iii) $u = 0$ at $y = 0$	(iv) $u = 0$ at $y = 3$

We can now narrow down some of the constants in our solutions. Starting with $X(x)$ from equation (17), we can see that $B_1 = 0$ since (i) tells us the cosine term must disappear at $x = 0$. Similarly, (ii) implies the sine term should disappear at $x = 3$, which means its argument should be an integer multiple of π. Thus, $\frac{b}{v^2} \cdot 3 = n\pi$, so $b = \frac{v^2\pi}{3}n$ for some integer n. Now we turn our attention to $Y(y)$ in equation (18). Here, $C_1 = 0$ again so that the cosine can disappear to satisfy (iii). Also, since (iv) says the sine term should be zero at $y = 3$, we get $c = \frac{v^2\pi}{3}m$, where m is an integer.

These frequencies $b = \frac{v^2\pi}{3}n$ and $c = \frac{v^2\pi}{3}m$ define the infinite set of "normal modes" I spoke of while introducing the wave equation. From these solutions we can build up our particular solutions by finding the proper coefficients. Plugging in all of our new constants to the solution $T(t)Y(y)X(x)$, our full form solution is now a sum over positive integers n and m. Combining the old constants as $B_2C_1A_1 = A$ and $B_2C_1A_2 = B$, we can write the sum out as:

$$u(x,y,t) = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty}\Big[Asin\Big(\frac{\pi}{3}nx\Big)sin\Big(\frac{\pi}{3}my\Big)cos\Big(\frac{v^2\pi}{3}(n+m)t\Big)$$

$$+Bsin\Big(\frac{\pi}{3}nx\Big)sin\Big(\frac{\pi}{3}my\Big)sin\Big(\frac{v^2\pi}{3}(n+m)t\Big)\Big] \qquad (19)$$

Part D: The Fourier Trick

We only have two constants left and we can find these using our last initial condition, given by $u(x,y,0)$ from equation (6) and by the initial speed $u'(x,y,0) = 0$. We will start by finding the constant A. The initial position at $t = 0$ cannot help us find the constant B since a factor of $sin(\frac{v^2\pi}{3}(n+m)t)$ will cause the entire second term from (19) to be zero at $t = 0$. By plugging in $u(x,y,0)$, we do find that (19) reduces to

$$u(x,y,0) = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty}Asin\Big(\frac{\pi}{3}nx\Big)sin\Big(\frac{\pi}{3}my\Big) \qquad (20)$$

To find the coefficient A, we will perform what is known as the *Fourier trick*. For this, multiply (20) through by $sin(\frac{\pi}{3}m'y)sin(\frac{\pi}{3}n'x)$ and integrate with respect to x and y over [0,3]. We then get

$$\Big(\frac{2}{3}\Big)^2\int_0^3\int_0^3 u(x,y,0)sin\Big(\frac{\pi}{3}m'y\Big)sin\Big(\frac{\pi}{3}n'x\Big)dxdy$$

$$= \sum_{n,m=1}^{\infty}A\Big(\frac{2}{3}\Big)^2\int_0^3\int_0^3 sin\Big(\frac{\pi}{3}nx\Big)sin\Big(\frac{\pi}{3}my\Big)sin\Big(\frac{\pi}{3}m'y\Big)sin\Big(\frac{\pi}{3}n'x\Big)dxdy \qquad (21)$$

On the right hand side, all the terms in the series with $n \neq n'$ and $m \neq m'$ drop out of the sum and each of the well-known integrals evaluate to $(\frac{3}{2})$. Thus we are left with $A(\frac{2}{3})^2(\frac{3}{2})^2$ on the right side of equation (21). Replacing our only remaining indices n', m' with n, m, we can now isolate A to find

$$A = \Big(\frac{2}{3}\Big)^2\int_0^3\int_0^3 u(x,y,0)sin\Big(\frac{\pi}{3}my\Big)sin\Big(\frac{\pi}{3}nx\Big)dxdy \qquad (22)$$

We could carry out the same process to find B from our remaining initial condition. However, since $u'(x,y,0) = 0$, we would end up with a similar integral as (22) but over an integrand of zero. This result would tell us that the coefficient $B = 0$, but this conclusion is fairly obvious without further derivation.

The double integral in (22) is nontrivial. In fact, plugging this into a powerful symbolic integral calculator like SymPy or Wolfram Alpha should discourage the reader from attempting to solve it by hand, as these too shall fail at finding an analytical solution. Instead, we will numerically compute the integral to find each coefficient using SciPy's quad function. The approximate values for the first four coefficients $A_{n,m}$ are given by [2.57, 1.70, 1.95e-6, 0.33].

Part E: Plotting The Results

With a value for each $A_{n,m}$ obtained through numeric integration, we can now plot our solution overtime. We will truncate the infinite series at $n, m = 4$, so our solution takes the following form:

$$u(x, y, t) = \sum_{n=1}^{4} \sum_{m=1}^{4} A_{n,m} sin\Big(\frac{\pi}{3}nx\Big) sin\Big(\frac{\pi}{3}my\Big) cos\Big(\frac{v^2\pi}{3}(n+m)t\Big). \tag{23}$$

We can now plot the height of the surface in 2D at a given time t. Before we watch the wave evolve in time, we should check to see that we have included enough terms in the series for a decent approximation. To do this, we will plot the initial Gaussian perturbation alongside our solution at $t = 0$. These should look more similar as we include more terms in the sum above $n, m = 4$. This described improvement can be seen in Fig. 2, where we find that including up to four terms gives a pretty good approximation without taking too long for the code to run. Finally, we can plot the wave over a range of time within 10 seconds. This evolution is shown in a series of plots on Fig. 3, where we see the transverse wave behavior that we expect.

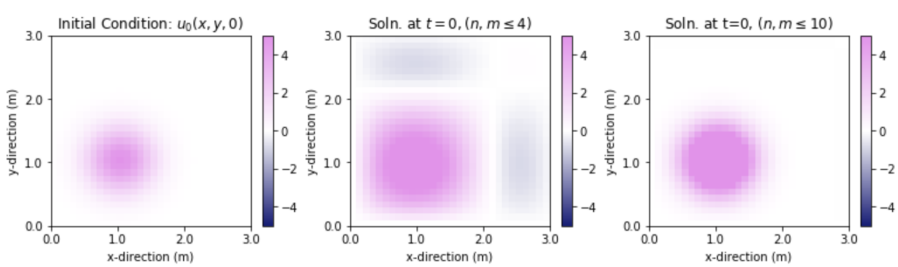

Figure 2: These plots show how the numeric solution (center and right plots) approach the actual initial conditions (plotted left). If we include 10 terms in the series (plotted right) instead of just 4 terms (plotted center), we can better approximate the displacement at $t = 0$.

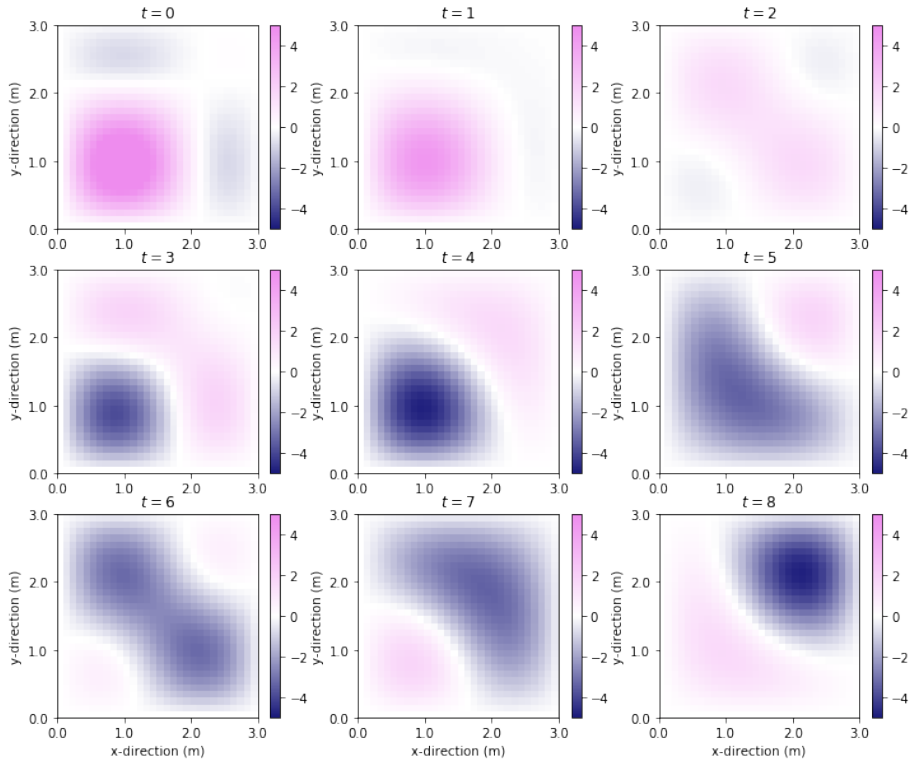

Figure 3: This collection of plots shows the perturbed surface's evolution over time in range [0,8] seconds. As expected for transverse waves, the wave peak moves in the xy-plane within the boundary causing troughs to form in other regions.